

Tao Ma

CONTACT INFORMATION	 xiaobaobaochifan.github.io  tao.ma@ucl.ac.uk	 linkedin.com/in/tao-ma-93642013a  github.com/xiaobaobaochifan
ACADEMIC POSITION	University College London (UCL) Global Business School for Health Lecturer (Assistant Professor) in Health Informatics and Statistics	Apr 2026 – Present London
	London School of Economics and Political Science (LSE) Research Assistant Mentor: Prof. Qiwei Yao	Oct 2025 – Dec 2025 London
EDUCATION	London School of Economics and Political Science (LSE) Doctor of Philosophy in Statistics Advisor: Prof. Zoltán Szabó	Sept 2021 – Sept 2025 London
	New York University (NYU) Courant Institute Master of Science in Mathematics	Sept 2016 – Aug 2018 New York
	Hong Kong Baptist University (HKBU) Bachelor of Science in Statistics and Operations Research (Science Elite Program)	Sept 2012 – Aug 2016 Hong Kong
RESEARCH INTERESTS	• Theory & methodology: reinforcement learning, large language models, deep generative models. • Trustworthy & interpretable AI: sustainability, sparse autoencoders, offline learning, causality. • Applications: healthcare, supply chains, organizational change, time-series forecasting, robotics. • AI behaviours: persuasion, sycophancy, deliberation, policy, social risks.	
PUBLICATIONS AND PREPRINTS	Ma, T., Zhu, J., Cai, H., Qi, Z., Chen, Y., Shi, C., & Laber, E. B. (2026). Sequential Knockoffs for Variable Selection in Reinforcement Learning. Journal of the American Statistical Association (JASA).  Schoenegger, P., ..., Ma, T., et al. (2025+). When Large Language Models are More Persuasive Than Incentivized Humans, and Why.   Ghilardi, D., Belotti, F., Molinari, M., Ma, T., & Palmonari, M. (2025). Group-SAE: Efficient Training of Sparse Autoencoders for Large Language Models via Layer Groups. EMNLP.  Ma, T., Yang, X., & Szabó, Z. (2025+). To Switch or not to Switch? Balanced Policy Switching in Offline Reinforcement Learning.  Ma, T., & Yang, X. (2024). A Framework for Policy Evaluation Enhancement by Diffusion Models. Tiny Papers ICLR. (Invite to present)	
SELECTED ACHIEVEMENTS	• LSE-NYU Research Seed Fund • LSE PhD Studentship • LSE Class Teacher Award • Best TA Teaching Award (HKUST) • HKSAR Government Scholarship (highest honor for undergraduates in Hong Kong) • Outstanding Student Award (7/4134, by The Baptist Convention of Hong Kong)	2025 – 2026 2021 – 2025 2022 – 2023 2019 – 2020 2013 – 2016 2015 – 2016

	• Scholastic Award 2015 – 2016 • APEC Scholarship (by Asia Pacific Economic Cooperation) 2014 – 2016 • Best Debater prize at The World Mandarin Debating Championship, Hong Kong 2015 – 2016 • Summer Undergraduate Research Fellowship 2014 – 2015 • Mr. Li Men Jan Prize in Mathematics 2014 – 2015 • Department of Mathematics Outstanding Academic Performance Award 2013 – 2014 • Sir Tseng Chi Lu Scholarship 2013 – 2014 • President’s Honour Roll 2012 – 2016
SKILLS	Python, R, SAS, MATLAB, $\text{\LaTeX}$, HTML, CSS, JavaScript, LEAN, Bash, PyTorch, VS Code, Linux, Git, Amazon Web Services, Google Cloud Platform, local HPC management
TEACHING EXPERIENCE	University College London London <i>Lecturer/Module Co-Lead</i> • Undergraduate courses: ◇ GBSH0050 Introduction to Statistics and Data Insights Jan 2027 – Mar 2027 - Introductory statistics and machine learning • Graduate courses: ◇ GBSH0009 Research Project Apr 2026 – Present - Research project course for Master’s students ◇ CHME0016 Artificial Intelligence in Healthcare Jan 2027 – Mar 2027 - Machine learning algorithms and AI methods for healthcare with Python London School of Economics and Political Science London <i>Class Teacher/Teaching Assistant</i> • Undergraduate courses: ◇ ST101 Programming for Data Science (Python) Sept 2022 – Dec 2024 - NumPy, Pandas, testing, object-oriented programming, functional programming ◇ ST107 Quantitative Methods Jan 2022 – May 2022 - Introductory statistics for students in social science ◇ ST102/109/110 Elementary Statistical Theory Sept 2022 – May 2025 - One-year course in probability and statistics • Graduate courses: ◇ ST445 Managing & Visualizing Data (Python) Sept 2021 – Dec 2021 - Pandas, BeautifulSoup, Sklearn, Jupyter Notebook Hong Kong University of Science and Technology (HKUST) Hong Kong <i>Research Assistant</i> Oct 2018 – Nov 2019 ◇ Mathematical physics, partial differential equations, signal processing <i>Class Teacher</i> Feb 2020 – May 2020 ◇ MATH 2011 Introduction to Multivariable Calculus New York University New York <i>Recitation Leader</i> Sept 2017 – Dec 2017 ◇ MATH-UA 211 Mathematics for Economics

ACADEMIC SERVICE	Technology Innovation and Data-Informed Healthcare Symposium	Sept 2026 – Oct 2026
	<i>Reviewer Panel</i>	
	International Conference on Econometrics and Statistics (EcoSta 2026)	Aug 2026
	<i>Session Chair</i>	
	LSE Statistics Research Showcase	Apr 2025
	<i>Session Chair</i>	
	Neural Information Processing Systems (NeurIPS)	May – Sept 2024 – 2026
	<i>Reviewer</i>	
	International Conference on Machine Learning (ICML)	Feb 2025 – May 2025
	<i>Reviewer</i>	
	International Conference on AI and Statistics (AISTATS)	Oct – Jan 2024 – 2026
	<i>Reviewer</i>	
	International Conference on Learning Representations (ICLR)	Oct – Jan 2024 – 2027
	<i>Reviewer</i>	
	London School of Economics and Political Science	London
	<i>Committee member</i>	
	• Department Review Representative	2024
	• Department Teaching Committee	2023 – 2025
	• Department Moving Plan Committee	2023 – 2025
	IMS Annual Meeting	June 2022
PROFESSIONAL EXPERIENCE	<i>Sessions Coordinator</i>	London
	University Mandarin Debate Team	Sept 2012 – Aug 2016
	<i>President, coach</i>	Hong Kong
	• Organized weekly training sessions, managed administrative tasks, participated in international competitions and gave 3-round interviews to around 100 applicants each year.	
	Professional Swimming	2004 – 2009
	Individual Medley (200 meters), Breaststroke (100 meters), Butterfly Stroke (100 meters)	Beijing
	Invited Talks	
	Balanced Policy Switching in Reinforcement Learning.	Aug 2026
	The 9th International Conference on Econometrics and Statistics (EcoSta 2026), Kyoto, Japan.	
	Balanced Policy Switching in Reinforcement Learning.	May 2025
INVITED TALKS	Workshop at Representing, calibrating & leveraging prediction uncertainty from statistics to machine learning, Isaac Newton Institute & University of Cambridge, UK.	
	Balanced Policy Switching in Reinforcement Learning.	Apr 2025
	Statistics Research Showcase, LSE, UK.	
	Careers in Data Science and AI.	Feb 2024
	The Alan Turing Institute & The Brilliant Club, UK.	
	Sequential Knockoffs for Variable Selection in Reinforcement Learning.	May 2023
	PhD Presentation Events, LSE, UK.	
	Sequential Knockoffs for Variable Selection in Reinforcement Learning.	Apr 2023
	Prof. Tengyao Wang's Reading Group, LSE, UK.	
	Sequential Knockoffs for Variable Selection in Reinforcement Learning.	Mar 2023
	Prof. Qiwei Yao's Reading Group, LSE, UK.	

Sequential Knockoffs for Variable Selection in Reinforcement Learning. RL+X Online Seminar.	Feb 2023
Reinforcement Learning with Representation Learning. LSE PhD Reading Group, UK.	Oct 2022
Nonstationarity in a Markov Decision Process. RL+X Online Seminar.	June 2022
Sequential Knockoffs for Variable Selection in Reinforcement Learning. Statistics Research Showcase (Poster Section), LSE, UK.	June 2022

LANGUAGES	Mandarin (native), English (fluent), Cantonese (fluent)
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GAMES	Poker, Mahjong, Pokémon, Mario, Kirby, Dragon Ball, NBA 2K, Asphalt, FIFA, Horizon 4, Zelda, GTA 5/6 and a lot of others
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